

02_ER Diagram & Database Schema

ER Diagram Overview

An Entity Relationship (ER) Diagram is used to visually represent how tables within the database are connected through primary and foreign key relationships. It helps understand the relational structure of the dataset and supports database design, analytical workflows, and business analysis.

Core Relationships

Relationship	Type
customers → orders	One-to-Many
orders → order_items	One-to-Many
products → order_items	One-to-Many
sellers → order_items	One-to-Many
orders → payments	One-to-Many
orders → reviews	One-to-One / One-to-Many
products → category_translation	Many-to-One

Important Understanding

Parent Tables

Contain master/descriptive information.

Examples:

- customers
- products
- sellers

Child Tables

Contain transactional or connected records.

Examples:

- orders
- order_items
- payments

Schema Understanding

The dataset follows a relational schema where transactional tables are connected to descriptive tables through shared identifiers. This structure supports customer, revenue, product, operational, and customer experience analysis.

Important Relationships Identified

- customer_id connects customers and orders
- order_id connects orders with order_items, payments, and reviews
- product_id connects products and order_items
- seller_id connects sellers and order_items
- product_category_name connects products and category_translation

The following ER diagram represents the relational structure of the Olist e-commerce dataset. It visualizes how transactional and descriptive tables are connected through primary and foreign key relationships.

